

Wachirawit Piyaprapapan

Data Scientist | Applied AI Engineer

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Education

Chulalongkorn University, B.Eng. in Electrical Engineering – Bangkok, Thailand

Aug 2022 – May 2026

- GPAX: 3.48 / 4.00; Second-class honours.
- Relevant coursework: Data Science, Data Engineering, Estimation, Statistical Learning, Optimization.
- Capstone: Led development of a generative video encoder-decoder forecasting model for Thai sky imagery, improving validation PSNR from 31.34 dB to 37.32 dB while prioritizing perceptual image quality.

Skills

Programming: Python, SQL, Bash

Machine Learning: Scikit-learn, MLForecast, AutoGluon, PyTorch, Hugging Face, Ollama, vLLM

Data & Visualization: Pandas, PySpark, PostgreSQL, Matplotlib, Seaborn, OpenCV

Deployment & MLOps: FastAPI, Docker, Git, GCP, Grafana, Prometheus, Airflow, dbt

Experience

Data Scientist Intern, AXONS – Bangkok, Thailand

June 2026 – Aug 2026

- Initiated an end-to-end demand forecasting pipeline across >2M series for logistics planning, replacing proxy-based estimates with direct shipment-weight forecasts and reducing forecast error by >50% and bias by >80% versus the existing baseline.
- Redesigned the evaluation set to reflect real-world demand variation and engineered special-day features, reducing forecast error by an additional 2.1% versus the best existing model.
- Reduced false zero-demand forecasts, improving forecast reliability for warehouse operations and vehicle-capacity planning on days previously predicted to have no demand.

AI Engineer, Part-time, Hobbit Technologies – Bangkok, Thailand

Feb 2026 – May 2026

- Designed and launched an internal AI-assisted English assessment platform for workplace-specific proficiency evaluation, reducing reliance on external learning tools and enabling personalized employee feedback.
- Built LLM-driven workflows for question generation, evaluation-based response scoring, and structured feedback, enabling more scalable and consistent employee assessment.
- Developed adaptive learning logic using learner performance and assessment history to personalize practice difficulty, improve progression tracking, and better target individual skill gaps.

AI Engineer Intern, Hobbit Technologies – Bangkok, Thailand

June 2025 – Aug 2025

- Led development and production deployment of an Edge AI vision system for real-time product detection, orientation classification, and counting under constrained on-device compute.
- Led fine-tuning and optimized a YOLO-based inference pipeline for edge deployment, achieving <5% misclassification while balancing model accuracy, latency, and compute constraints in production.
- Initiated integration of automated visual inspection and monitoring into the manufacturing line, enabling earlier detection of misoriented or misclassified parts and reducing downstream operational risk.

Electrical Engineering Intern, AGC Flat Glass – Bangkok, Thailand

June 2024 – Aug 2024

- Proposed and developed a data-driven conveyor control strategy, replacing a fixed PLC cycle runtime with dynamic timing based on production conditions.
- Collaborated with senior engineers to integrate the control logic into PLC workflows, reducing average conveyor cycle time and reducing operational runtime by >45%.

Projects

Generative Video-Based Sky Image Forecasting for Thai Sky Images

Aug 2025 – May 2026

- Led development of a generative video encoder-decoder forecasting model for Thai sky imagery, improving validation PSNR from 31.34 dB to 37.32 dB while prioritizing perceptual image quality.
- Revisited and benchmarked prior VQ-VAE architecture choices on the same sky dataset, then redesigned the forecasting pipeline around a pretrained VQ-VAE2 representation with Transformers and PhyCell for 15-minute stochastic cloud-motion prediction.
- Engineered a computer-vision data pipeline for 400–500 GB of sky video, reducing storage by ~95% through synchronized frame sampling, resizing, and batch serialization.

Football Player Value Forecasting & Similarity Recommendation System

Oct 2025 – Mar 2026

- Conceived and built an end-to-end football scouting analytics system modeled on a professional recruitment workflow, spanning player identification, performance profiling, comparable-player discovery, and market valuation.
- Reconciled 98.3% of cross-source player records and engineered position-aware per-90 profiles for 3,755 players, enabling consistent comparison across leagues, positions, and playing roles.
- Guided modeling strategy for player valuation, working with teammates to refine features and model design and reduce RMSE from 136% to 39% versus baseline, improving the reliability of transfer-value signals used for player shortlisting.
- Advised on player-similarity modeling and feature design to surface comparable transfer targets, supporting squad-fit assessment and succession planning.

Competitions

Microsoft AI Engineering Skills and Hackathon for Employment 2026 (Finalist)

2026

- Secured a finalist team spot from a competitive pool of 2,000+ regional participants.
- Designed and built a real-time emergency triage dashboard that transformed incoming cases into prioritized, actionable views for rapid response and case monitoring.

Thailand Super AI Engineer Season 6 — Edge AI for Intelligent Wellness Systems (Winner)

2026

- Invented a gas-leak detection system without conventional gas sensors, using infrared and ultrasonic propagation signals and achieving 0.76–0.92 recall across target gases.
- Led development of a worker fall-detection system, explicitly prioritizing recall to capture safety-critical events such as dizziness and falls, achieving 0.84 recall for preventive intervention.
- Built a real-time helmet safety monitoring system combining danger alerts with worker location tracking, reducing response delays to supervisors under 5 seconds.

Thailand Super AI Engineer Season 6 — Enterprise Data Agentic Showdown (First Runner-Up)

2026

- Architected an end-to-end financial document intelligence system spanning OCR, indexing, retrieval, and agentic question answering for complex financial documents.
- Developed an OCR pipeline using vLLM and open-weight models, achieving 91.4% accuracy on financial-document extraction tasks.
- Built a LangGraph-based OCR/QA agent that orchestrated retrieval, tool use, and self-correction, achieving 62% accuracy on the private evaluation set.

Thailand Super AI Engineer Season 6 — Coffee Chain Time-Series Hackathon (Pitching Team)

2026

- Challenged and replaced the existing retail demand-forecasting baseline with a LightGBM pipeline, reducing cross-validation MAE by 30.2% (13.150 → 9.176).
- Prioritized high-error, high-volume product lines such as coffee and feature engineering to improve forecasts where errors carried the greatest inventory-planning impact.
- Pitched the forecasting solution, model evidence, and inventory-planning value to stakeholders.

Thailand Super AI Engineer Season 6 — Computer Vision Hackathon

2026

- Researched and evaluated a multi-head ConvNeXt architecture for complex re-identification, exploring shared and task-specific representations across prediction heads.

I-squared Hackathon — Motorbike Rider Anomaly Detection & Classification (Semi-finalist)

2024

- Developed a Vision Transformer pipeline for safety-oriented motorbike anomaly classification, achieving 98.02% accuracy and 0.99 AUC.
- Optimized the classification workflow for high-separability detection of dangerous rider or vehicle patterns, supporting automated traffic-safety screening.

Volunteer Experience

Chulalongkorn University Electrical Engineering - Sustainable Development Goals – Saraburi, Thailand

March 2024 to March 2026

- Served for two years as technical staff, planning and implementing the camp's electrical infrastructure, including power distribution, equipment requirements, and on-site installation.
- Elected Camp President in the final year; restructured the organizing team and led end-to-end planning across school selection, activity design, procurement, logistics, staffing, and on-site execution.
- Redesigned responsibilities and operating workflows across the organization to improve coordination and execution across camp functions.